Visible Mismatch Vanishes While Causal Inventory Diverges

A parity-sensitive separation theorem for exact zero-repair information trees

Jan Mikulík

25 August 2026

Abstract

Consider the dyadic information-increment pair

$$\mu_P = \tfrac{1}{4}\delta_2 + \tfrac{3}{4}\delta_4, \qquad \mu_Q = \tfrac{3}{4}\delta_3 + \tfrac{1}{4}\delta_5.$$

The two laws have the same mean and variance but opposite third centered cumulants. A companion zero-repair theorem shows that the minimum initial nonnegative causal inventory required to absorb the mismatch exactly for n steps satisfies

$$\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{1}{3}.$$

Here we show that the corresponding *static endpoint mismatch* behaves in the opposite way. If

$$\delta_n := d_{\mathrm{TV}}(\mu_P^{*n}, \mu_Q^{*n}),$$

then $\delta_n = 1$ for every odd n, while along the even subsequence

$$\boxed{\delta_n = \frac{\sqrt{2/\pi}}{3\sqrt{3}}\left(1 + 4e^{-3/2}\right) n^{-1/2} + o(n^{-1/2}).}$$

In particular, on dyadic scales $n = 2^k$,

$$\delta_{2^k} = C_{\mathrm{stat}}\, 2^{-k/2} + o(2^{-k/2}), \qquad \mathcal{B}_{2^k} \geq \frac{\log 2}{3}\, k - o(k).$$

Thus the scalar endpoint discrepancy becomes exponentially small in renormalization depth while the exact causal working inventory grows at least linearly in that depth. The same third-order defect therefore produces vanishing static distinguishability and divergent causal inventory. This gives a precise separation between endpoint information profiles and prefix-compatible causal accounting, and independently explains why scale doubling is the natural renormalization in the parity example.

The endpoint forgets the path. The inventory cannot.

1 Two resource questions

There are two different ways to ask how far μ_P and μ_Q are from one another after n steps.

The first is a static endpoint question. If one is allowed to couple only the final accumulated information values, the smallest probability of disagreement is the total variation distance

$$\delta_n := \inf\{\mathbb{P}(S_P \neq S_Q) : S_P \sim \mu_P^{*n},\ S_Q \sim \mu_Q^{*n}\} = d_{\mathrm{TV}}(\mu_P^{*n}, \mu_Q^{*n}). \tag{1}$$

The second is the exact zero-repair question of the companion work. A history-indexed family of nonnegative buffer laws must satisfy at every internal history

$$\mu_Q * B_h = \frac{1}{4}\delta_2 * B_{h0} + \frac{3}{4}\delta_4 * B_{h1}, \tag{2}$$

and $\mathcal{B}_n$ is the infimum of the root mean $\mathbb{E}J$ over depth-n feasible trees. The established converse is

$$\boxed{\liminf_{n\to\infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{1}{3}.} \tag{3}$$

The quantity δ_n asks only whether the two *final scalar totals* can be coupled. The quantity $\mathcal{B}_n$ asks whether the discrepancy can be carried causally and nonnegatively through every prefix with no fresh repair. They need not have the same asymptotics.

2 Exact parity and reflection

Write

$$X_i = 2 + 2\xi_i, \qquad \xi_i \sim \text{Bernoulli}(3/4),$$

so that $X_i \sim \mu_P$, and

$$Y_i = 3 + 2\eta_i, \qquad \eta_i \sim \text{Bernoulli}(1/4),$$

so that $Y_i \sim \mu_Q$.

Proposition 2.1 (Parity obstruction)**.** *For every odd n,*

$$\boxed{\delta_n = 1.}$$

Proof. Every sum of n P-increments is even. Every Q-increment is odd, so a sum of an odd number of them is odd. Hence the supports of μ_P^{*n} and μ_Q^{*n} are disjoint. □

The even subsequence is completely different.

Proposition 2.2 (Exact reflection on even blocks)**.** *Let $n = 2r$. If K, J are independent copies of* $\text{Bin}(n, 3/4)$*, then*

$$S_P \overset{d}{=} 2n + 2K, \qquad S_Q \overset{d}{=} 5n - 2J.$$

Both have mean $7n/2$, and

$$\boxed{S_Q - \frac{7n}{2} \overset{d}{=} -\left(S_P - \frac{7n}{2}\right).} \tag{4}$$

Thus, for even n, the Q endpoint law is the reflection of the P endpoint law about their common mean.

Proof. The first identity is immediate from $S_P = 2n + 2\sum_i \xi_i$. Put $J = n - \sum_i \eta_i$. Since $1 - \eta_i \sim \text{Bernoulli}(3/4)$, one has $J \sim \text{Bin}(n, 3/4)$, and

$$S_Q = 3n + 2(n - J) = 5n - 2J.$$

The common mean is $7n/2$, and subtraction gives (4). □

Remark 2.3 (Why doubling is structural)*.* The sequence δ_n has no ordinary smooth large-n behavior across all integers: it equals one on every odd block and tends to zero on even blocks. Passing from n to $2n$ is therefore not merely a convenient choice suggested by generating functions. It is the minimal scale change that remains inside the lattice-aligned sector where the cubic defect can be seen after the parity obstruction has been factored out.

3 Even-block asymptotics

Let $p = 3/4$, $q = 1/4$, and put

$$\sigma_B^2 = pq = \frac{3}{16}, \qquad \lambda_3 := \frac{pq(q-p)}{(pq)^{3/2}} = \frac{q-p}{\sqrt{pq}} = -\frac{2}{\sqrt{3}}.$$

For $K \sim \mathrm{Bin}(n, p)$ define

$$Z_n := \frac{K - np}{\sigma_B \sqrt{n}}.$$

The standard local Edgeworth expansion for a lattice sum with finite moments gives, uniformly in the central range,

$$\mathbb{P}(K = k) = \frac{1}{\sigma_B \sqrt{n}} \varphi(x) \left[1 + \frac{\lambda_3}{6\sqrt{n}} H_3(x)\right] + o(n^{-1}), \tag{5}$$

where

$$x = \frac{k - np}{\sigma_B \sqrt{n}}, \qquad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \qquad H_3(x) = x^3 - 3x.$$

A standard exponential tail bound allows the expansion to be summed in total variation; see, for example, Petrov's treatment of asymptotic and local limit expansions for lattice sums [1].

Theorem 3.1 (Static cubic mismatch)**.** *Along even n,*

$$\boxed{\delta_n = C_{\mathrm{stat}}\, n^{-1/2} + o(n^{-1/2}),} \tag{6}$$

where

$$\boxed{C_{\mathrm{stat}} = \frac{1}{3\sqrt{3}} \mathbb{E}|H_3(Z)| = \frac{\sqrt{2/\pi}}{3\sqrt{3}} (1 + 4e^{-3/2}) = 0.2906021373\ldots,} \tag{7}$$

with $Z \sim N(0, 1)$.

Proof. By Proposition 2.2, for even n the endpoint problem is the total variation distance between the law of $K \sim \mathrm{Bin}(n, 3/4)$ and its reflection about $np = 3n/4$. In standardized coordinates the reflected point has coordinate $-x$.

Subtracting (5) at x and $-x$, using that φ is even and H_3 is odd, gives

$$\mathbb{P}(K = k) - \mathbb{P}(K = k^\star) = \frac{1}{\sigma_B \sqrt{n}} \varphi(x) \frac{\lambda_3}{3\sqrt{n}} H_3(x) + o(n^{-1}),$$

uniformly on a central window whose complement has negligible total mass. Hence the Riemann-sum limit yields

$$\delta_n = \frac{1}{2} \sum_k |\mathbb{P}(K = k) - \mathbb{P}(K = k^\star)| = \frac{|\lambda_3|}{6\sqrt{n}} \int_{\mathbb{R}} |H_3(x)| \varphi(x)\, dx + o(n^{-1/2}).$$

Since $|\lambda_3| = 2/\sqrt{3}$, the first equality in (7) follows.

It remains only to evaluate the Gaussian integral. The zeros of H_3 are 0 and $\pm\sqrt{3}$. By symmetry and the identities

$$\int_a^\infty x\varphi(x)\, dx = \varphi(a), \qquad \int_a^\infty x^3 \varphi(x)\, dx = (a^2 + 2)\varphi(a),$$

one obtains

$$\int_{\mathbb{R}} |x^3 - 3x| \varphi(x)\, dx = 2\varphi(0) + 8\varphi(\sqrt{3}) = \sqrt{\frac{2}{\pi}} (1 + 4e^{-3/2}).$$

Substitution proves (7). ∎

Remark 3.2 (The same third cumulant). For the information increments themselves,

$$\sigma^2 = \frac{3}{4}, \qquad \kappa_3(P) = -\frac{3}{4}, \qquad \kappa_3(Q) = +\frac{3}{4}, \qquad \Delta\kappa_3 = \frac{3}{2}.$$

The general first-order Edgeworth coefficient for two lattice-aligned laws with matching mean and variance is

$$\frac{|\Delta\kappa_3|}{12\sigma^3}\mathbb{E}|H_3(Z)|.$$

For the present pair this reduces exactly to (7). Thus the same third-order defect that drives the zero-repair converse also controls the leading static endpoint separation, but the two resource laws have opposite asymptotic directions.

4 Visible–causal separation on dyadic scales

Combining Theorem 3.1 with the established zero-repair lower bound (3) gives the main consequence.

Theorem 4.1 (Visible–causal separation). *Let $n = 2^k$ with $k \to \infty$. Then*

$$\boxed{\delta_{2^k} = C_{\text{stat}}\, 2^{-k/2} + o(2^{-k/2}),} \tag{8}$$

$$\boxed{\mathcal{B}_{2^k} \geq \frac{\log 2}{3}\, k - o(k).} \tag{9}$$

Consequently,

$$\boxed{\delta_{2^k} \longrightarrow 0 \qquad \textit{while} \qquad \mathcal{B}_{2^k} \longrightarrow \infty.} \tag{10}$$

Proof. Equation (8) is (6) with $n = 2^k$. Equation (9) follows from (3) because

$$\log(2^k) = k \log 2.$$

□

The theorem isolates a sharp failure of endpoint reasoning. At scale 2^k, the final scalar laws can be coupled with disagreement probability of order $2^{-k/2}$, yet an exact nonnegative history-indexed inventory must carry at least order k mean capital if fresh repair is forbidden.

Equivalently,

$$\boxed{\text{visible endpoint discrepancy decays exponentially in scale depth,}}$$

while

$$\boxed{\text{causal zero-repair inventory grows linearly in scale depth.}}$$

This is not a contradiction: the endpoint coupling is free to use information about the entire block, while the zero-repair tree must remain prefix-compatible at every history.

Corollary 4.2 (No endpoint-TV control of the causal bill). *There is no upper bound of the form*

$$\mathcal{B}_n \leq F\Big(d_{\text{TV}}(\mu_P^{*n}, \mu_Q^{*n})\Big)$$

along the dyadic subsequence for any function F that remains bounded in a neighborhood of zero.

Proof. Along $n = 2^k$, the argument of F tends to zero while $\mathcal{B}_n$ diverges. □

5 Exact finite values and the asymptotic constant

For orientation, the static mismatch on the first even blocks is

n	δ_n	$\sqrt{n}\,\delta_n$
2	1/4	0.353553
4	5/32	0.312500
6	293/2048	0.350440
8	821/8192	0.283464
10	49709/524288	0.299823

The scaled quantity converges to

$$C_{\text{stat}} = 0.2906021373\ldots.$$

The small-block oscillation is a lattice effect; Theorem 3.1 concerns the even asymptotic subsequence.

6 Interpretation for causal phase obstructions

The separation theorem gives an independent reason that the scalar information profile is too small a state for the sharp causal problem.

At the endpoint level, matched mean and variance force the first visible discrepancy into a third-order Edgeworth term, and the static mismatch therefore vanishes like $n^{-1/2}$ on aligned blocks. In the zero-repair problem, however, the same third-order defect is repeatedly exposed to positivity and prefix compatibility. The lower-bound mechanism converts that defect into a logarithmically growing inventory requirement.

The contrast is therefore not

$$\text{large mismatch versus small mismatch,}$$

but rather

$$\boxed{\text{vanishing scalar mismatch versus accumulating causal compatibility cost.}}$$

This is exactly the role played by the hidden ordering phase in the companion phase-geometry paper: scalarization removes information that is irrelevant to a terminal coupling but can remain decisive for an adapted one.

7 What is not proved

Theorem 4.1 should not be confused with a sharp asymptotic solution of zero repair.

- It does not prove $\mathcal{B}_n = O(\log n)$; only the existing logarithmic lower bound is used.
- The quantity δ_n is a *static endpoint* mismatch. It is not the causal mismatch LP studied at fixed block length, and it is not the unrestricted causal entropy residual.
- The parity identity is special to the present dyadic pair. The Edgeworth mechanism, by contrast, is the standard third-cumulant phenomenon on lattice-aligned subsequences.

- No historical novelty is claimed for the local Edgeworth expansion itself. The point is the resource separation obtained by placing that classical asymptotic next to the exact zero-repair converse.

The remaining sharp question is unchanged:

$$\boxed{\sup_n(\mathcal{B}_{2n}-\mathcal{B}_n)<\infty\ ?}$$

A positive answer would turn the lower bound into $\mathcal{B}_n=\Theta(\log n)$. The present note shows that such a theorem cannot be read off from the final scalar discrepancy: by the time the causal bill has become large, the visible endpoint mismatch has already become small.

8 Conclusion

The dyadic pair exhibits a two-layer critical phenomenon.

First, parity makes odd blocks perfectly distinguishable. Doubling removes this lattice obstruction. On the aligned even subsequence, the first two moments cancel and the third cumulant becomes the leading visible defect, producing

$$d_{\mathrm{TV}}(\mu_P^{*n},\mu_Q^{*n})\asymp n^{-1/2}.$$

Second, under exact nonnegative zero repair, the same third-order mismatch cannot simply be paid once at the endpoint. Prefix compatibility forces a growing working inventory:

$$\mathcal{B}_n\ge\left(\frac{1}{3}-o(1)\right)\log n.$$

Thus the endpoint becomes statistically indistinguishable at the same time that the causal accounting problem becomes more expensive.

$$\boxed{\text{The visible mismatch disappears. The causal bill does not.}}$$

Research-status note

The zero-repair lower bound cited here is proved in the companion manuscript [2]. The phase interpretation is developed in [3]. The only external asymptotic input in the new theorem is the standard local Edgeworth expansion for lattice sums. Computational values in the table are exact binomial calculations and are not used as proof.

References

[1] V. V. Petrov, *Sums of Independent Random Variables*, Ergebnisse der Mathematik und ihrer Grenzgebiete, Vol. 82, Springer-Verlag, Berlin–Heidelberg–New York, 1975.

[2] J. Mikulík, *Who Pays the Bill? Exact Zero-Repair Information Trees and a Third-Cumulant Logarithmic Obstruction*, research manuscript, 2026.

[3] J. Mikulík, *Causal Phase Obstructions in Exact Information Simulation: Zero-Repair Trees, Hidden Ordering Information, and Multiscale Renormalization*, research manuscript, 2026.